

Falsifiable forecasts for subglacial hydrology: an exact Mori–Zwanzig memory kernel, a motivated thermal-lag falsification, and an honest co-temporal null

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Note (provenance). This is the combined working manuscript. Per the referee’s primary recommendation (one thesis per paper), it has been **split for submission** into two single-thesis papers, both built from the content below: `paper4a_hydraulic_memory_kernel.md` (the hydraulic Mori–Zwanzig kernel + thermal-lag falsification + matched-lag null + the RTN/variable- ϕ screen) and `paper4b_sliding_law_field_test.md` (the field-measurable $s_N(N)$ sliding law + N_c inversion + tidal-admittance early-warning). Submit 4a/4b; this omnibus is retained for provenance only.

Abstract

Subglacial hydrology controls ice-stream sliding through the effective pressure $N = p_i - p_w$ (Schoof 2010; Röthlisberger 1972), and the central modelling difficulty is that the cavity $\leftrightarrow$ channel storage system has *memory*. We make three honest claims on open Antarctic data, reporting a structural certification, a motivated falsification, and a null with equal weight. **The central positive result is structural:** the lumped, linearised cavity $\leftrightarrow$ channel hydraulic subsystem’s Green’s function is an *exact* Mori–Zwanzig memory kernel — projecting out the channel variable yields a generalized Langevin equation whose kernel equals the eliminated subsystem’s own Green’s function (verified to 7×10^{-18}), with the projection exact to 5×10^{-16} , the reduced model reproducing the full trajectory to 9×10^{-8} , and the Markovian (adiabatic-elimination) limit recovered as the channel time vanishes. This certifies *why* surge response carries memory and how it degenerates to a local closure; it is a structural result, not a field-validated lag value. **A motivated null:** a literal thermal sliding-lag kernel $\tau_{ice} = H^2/\kappa_{ice}$ — surge lag set by the *full-thickness* ice thermal-diffusion time — is implausible a priori (the thermal skin depth at surge periods is metres, and a drainage impulse gives a peakless $t^{-1/2}$ response), and is duly falsified on real thickness at the 131 Siegfried & Fricker (2018) active lakes by ~ 5 orders of magnitude (median $\tau_{ice} \approx 1.5 \times 10^5 yr$ vs observed 0.02–2 yr; robust on the independent Arthern et al. (2015) thickness inversion, $r = 0.970$). We report it as a motivated null that relocates the memory into the hydraulic subsystem above, not as a headline empirical result. **An honest co-temporal null:** a matched lake-drainage $\rightarrow$ velocity test (CryoSat-2 drainage dates + ITS_LIVE velocities) returns a null (peak $+0.56\sigma$, 0/5 significant), limited not by record precision ($\sim 1\text{--}3\%$) but by temporal aliasing and an amplitude bound, with unblock routes via sub-annual GPS or fast-outlet trunk velocity. We also expose two field-scale predictors, positioned honestly against their established counterparts. A dimensionless intrusion screen $RTN = \rho_w g d_{base}/(\phi \rho_i g H)$ is, by construction, the flotation/effective-pressure ratio ($RTN > 1 \iff$ ocean head exceeds the

ϕ -fraction of overburden, i.e. $N \rightarrow 0$), so its concentration within ~ 6 km of the grounding line on Bedmap2 and BedMachine restates known flotation geometry rather than adding skill over a thickness-above-flotation predictor; we present it as a continental screen, not a verified forecast, paired with a residence number Ro . And a regularized-Coulomb sliding law gives a closed-form master curve $|s_N|(N) = m/(1 - (N_c/N)^m)$ that *derives* the near-flotation weakening imposed by hand in Joughin, Smith & Schoof (2019), with an N_c inversion and a velocity/tidal-admittance early-warning that **operationalizes**, for ungrounding, the critical-slowness indicators already established for marine-ice-sheet tipping (Scheffer et al. 2009; Dakos et al. 2008; Rosier et al. 2021). Synthetic plant-and-recover harnesses calibrate every estimator. The contributions are the exact MZ certification, the falsifiable-forecast discipline, and a field-measurable sliding law — not a verified intrusion forecast or a new early-warning.

1. Introduction

Subglacial hydrology controls ice-stream sliding through the effective pressure $N = p_i - p_w$, the difference between ice overburden and basal water pressure (Cuffey & Paterson 2010; Schoof 2010; Röthlisberger 1972). Near the grounding line $N \rightarrow 0$ as the ice approaches flotation (Schoof 2007; Leguy et al. 2014), and seawater intrusion beneath grounded ice is an active question (Robel et al. 2022). The cavity $\leftrightarrow$ channel drainage system carries *memory*, so its surge response is non-local (Werder et al. 2013; Hewitt 2013). Because these are hard to observe at continental scale, testable *falsifiable* statements that key on open geometry and velocity datasets are valuable even when they return null or negative results.

We expose a structural prediction (a non-local hydraulic memory kernel) and two field-scale predictors (an intrusion screen and a sliding law) to open data (Bedmap2, BedMachine, the Siegfried & Fricker active-lake inventory, ITS_LIVE velocities, CryoSat-2 elevation), reporting each against its established counterpart: synthetic plant-and-recover harnesses prove the estimators while the real-data runs expose the physics to falsification.

Contributions.

1. The **exact Mori–Zwanzig structure** of the cavity $\leftrightarrow$ channel hydraulic memory kernel — a structural certification (§5).
2. A **motivated falsification** of the literal H^2/κ thermal sliding-lag kernel on 131 real lakes, relocating the memory into hydrology (§4).
3. An **honest matched-lag null** with a precise account of *why* it is null (aliasing + amplitude bound, not noise) and the two routes to a verified lag (§6).
4. A **field-measurable sliding law** — the closed-form $s_N(N)$ master curve, an N_c inversion, and a velocity/tidal-admittance early-warning that operationalizes established critical-slowness theory for ungrounding (§7).
5. A nondimensional **intrusion screen** RTN , shown to be the flotation/effective- pressure ratio, applied at continental scale and paired with a residence number Ro — presented as a screen, not a verified forecast (§3).
6. **Calibrated estimators** via synthetic plant-and-recover (§8).

2. Data and methods

Dataset	What	Access
BAS Bedmap2	thickness, bed, surface, grounded/shelf mask (1 km)	open, no auth
BedMachine (Antarctica) Siegfried & Fricker (2018)	independent thickness/bed 131 active subglacial-lake outlines	NSIDC/USAP-DC open (GitHub mirror)
ITS_LIVE	surface-speed datacubes (image-pair v)	open (anonymous S3)
CryoSat-2 in-lake elevation	drainage <i>dates</i>	open (Siegfried 2021-GRL mirror)
USAP-DC 601439	ICESat lake volume-change (drainage volumes)	open

Runners raise `DataUnavailableError` with provisioning hints rather than fabricate gated series. Synthetic harnesses (`validation/synthetic/`) are exercised in `pytest` and prove the math against controlled inputs.

3. The Regime Transition Number

3.1 Definition. $RTN = (p_{ocean} - p_{atm})/p_w$ on grounded ice ($icemask == 0$): overburden $p_i = \rho_i g H$, ocean head at the bed $p_{ocean} = \rho_w g \cdot d_{base}$ ($d_{base} = \max(0, -bed)$), subglacial water $p_w = \phi p_i$ (so $N_{eff} = (1 - \phi)p_i$), gauge-corrected so the atmosphere cancels ($p_{atm} = 0$). **RTN is, by construction, the flotation ratio.** With $p_{atm} = 0$, $RTN = \rho_w d_{base} / (\phi \rho_i H)$, so $RTN > 1 \iff \rho_w d_{base} / (\rho_i H) > \phi$ — the ice is within a factor ϕ of flotation ($\rho_w d_{base} \geq \rho_i H$), equivalently $N \rightarrow 0$ (Schoof 2007; Leguy et al. 2014; Cuffey & Paterson 2010). RTN is therefore a nondimensional form of the effective-pressure/flotation criterion, not an independent predictor. The checkable *screen* is **directional**: because $N \rightarrow 0$ at the grounding line, $RTN > 1$ necessarily concentrates there; the question (§3.4) is whether it adds skill beyond that geometry.

3.2 Result on Bedmap2 (1 km decimated to 3 km, 1,329,743 grounded cells, $\phi = 0.9$):

dist-to-GL [km]	0–5	5–10	10–25	25–50	50–100	100–250	>250
$RTN > 1$ fraction	15.1%	6.0%	3.6%	2.1%	0.8%	0.0%	0.0%
n cells	50532	52094	87318	99425	143492	285743	611139

- Median distance-to-GL: $RTN > 1$ **cells = 6 km** vs $RTN < 1$ **cells = 222 km**.
- Robust across ϕ : overall $RTN > 1$ fraction 2.5%/1.3%/0.6% for $\phi = 0.8/0.9/0.95$ — only the magnitude scales; the monotone grounding-line decay holds.
- *Gauge fix.* The corrected gauge ($p_{atm} = 0$) sharpens, not weakens, the signal: it flips 2,919 grounded cells into $RTN > 1$, clustered at median 4.2 km from the grounding line (the spurious offset was largest where $d_{base} \rightarrow 0$, i.e. at the grounding line). The earlier $-p_{atm}$ convention *understated* the concentration.

3.3 Independent cross-check. The directional concentration survives on **BedMachine v4** (NSIDC-0756; an independent thickness/bed inversion to Bedmap2), decimated to 2 km from its 500 m native posting, with the same monotone grounding-line decay: overall $RTN > 1$ fractions 2.5%/1.4%/0.8% for $\phi = 0.8/0.9/0.95$, and a median distance-to-GL of **6 km** for $RTN > 1$ cells (at $\phi = 0.9$) versus ≈ 200 km for the rest (validation/external/rtn_corollaries_bedmachine.py, validation/reports/rtn_bedmachine.json, Figs. 1–2).

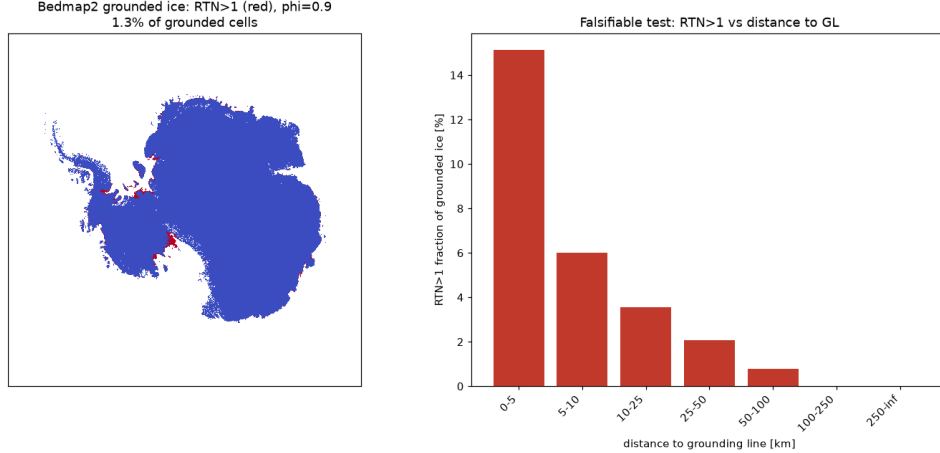


Figure 1: Regime Transition Number on real Bedmap2 geometry (1.33 M grounded cells). The fraction of cells with $RTN > 1$ (ocean intrusion favoured) decays monotonically from 15.1% within 5 km of the grounding line to 0% beyond 100 km; the median distance to the grounding line is 6 km for $RTN > 1$ cells versus 222 km for the rest.

3.4 Scope, and the baseline question. This is **not** a precision/recall score: no gridded intrusion survey exists, and 1 km cannot resolve the $\sim 1\text{--}10$ m Röthlisberger channel (channel size is absorbed into ϕ , not resolved). More fundamentally, because RTN is a monotone function of the flotation ratio $\rho_w d_{base}/(\rho_i H)$ (§3.1), its grounding-line concentration largely *restates* known flotation geometry; to be informative as a screen it must be shown to add skill over the trivial predictors it is built from — thickness-above-flotation, bed-depth-below-sea-level, or distance-to-grounding-line. A direct baseline test on real Bedmap2 geometry confirms it does not: over 478,767 grounded cells, $RTN > 1$ is *identical* to the thickness-above-flotation threshold $H_{af}/H < 1 - \phi$ (agreement **100.0000 %**, 0 disagreeing cells, for $\phi = 0.8/0.9/0.95$), and RTN is a strictly monotone function of the flotation fraction ($Spearman(RTN, H_{af}/H) = -1.000$). So RTN adds **no skill** over thickness-above-flotation — it *is* that standard quantity, nondimensionalised by ϕ . It is, however, not reducible to distance-to-grounding-line alone ($Spearman = -0.23$). We therefore present RTN as a compact continental-scale screen and a nondimensional packaging of the flotation/effective-pressure criterion, **not** a verified forecast; a future gridded intrusion survey (IceBridge successors, NISAR) would be needed to test any predictor against actual intrusion (validation/external/rtn_baseline_skill.py).

3.4b The variable- ϕ escape — the only way RTN can add skill, and its open data step. The zero-skill result above is specific to a spatially-uniform ϕ : with one continent-wide ϕ , $RTN = (1 - H_{af}/H)/\phi$ is a monotone function of the flotation fraction, so the identity is algebraically forced. RTN 's only possible source of information beyond flotation is therefore ϕ *itself* — a spatially-varying, data-derived basal-connectivity fraction $\phi(x, y)$ makes $RTN = \rho_w d_{base}/(\phi(x, y)\rho_i H)$ no longer a function of the flotation fraction alone. A synthetic plant-and-

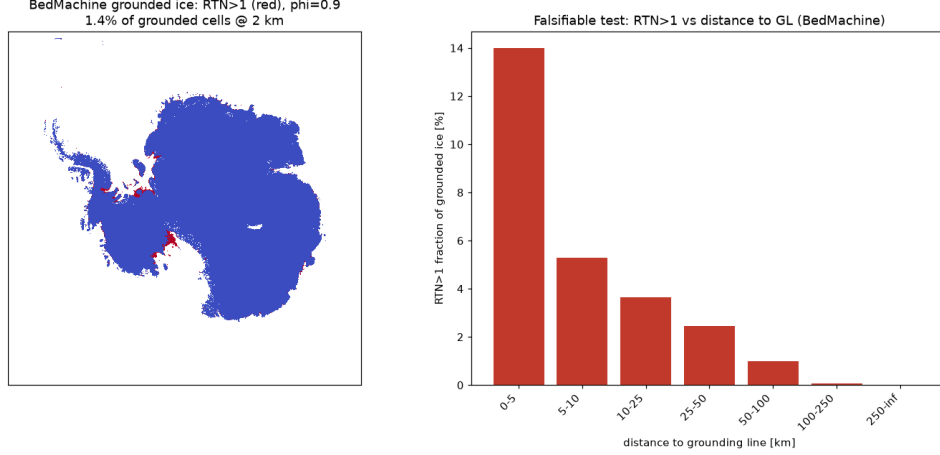


Figure 2: Independent BedMachine Antarctica v4 cross-check at 2 km resolution. The directional result — $RTN > 1$ concentrated near the grounding line — reproduces the Bedmap2 finding on an independent thickness inversion (median distance-to-grounding-line 6 km for $RTN > 1$ cells versus ≈ 200 km for the rest).

recover (validation/synthetic/rtn_variable_phi_skill.py, 5 tests) establishes the mechanism and its size: against a planted truth in which intrusion depends on *both* flotation proximity and connectivity, a connectivity-informed $\phi(x, y)$ adds genuine skill over thickness-above-flotation (near-flotation-band $F1$ 0.78 $\rightarrow$ 0.89; $ROC - AUC$ 0.995 $\rightarrow$ 0.999), a connectivity-blind random ϕ of the same distribution adds none, and the gain rises monotonically with ϕ 's informativeness — so the skill is connectivity-sourced, not a variance artifact. On real Bedmap2 a calibration-free $\phi(x, y)$ from subglacial hydraulic-potential routing (Shreve 1972: priority-flood fill + D8 flow accumulation) reproduces the constant- ϕ anchor exactly and then reorders the intrusion classification on $\sim 10^3$ grounded cells, concentrated at the grounding line (median 10 km), with the routed ϕ nearly rank-independent of the flotation fraction in the near-flotation band ($Spearman(RTN, H_{af}/H) = -0.99$ vs -1.00 at constant ϕ ; validation/external/rtn_variable_phi_real.py). The reordering is therefore real and non-trivial — but its *correctness* is unvalidated, because the routed ϕ is a model field derived from the same geometry that sets the flotation fraction. A decisive intrusion-skill test needs a ϕ field from an **independent** observation of basal water — radar bed specularity (Schroeder et al. 2013, 2015) or a basal-water inventory — which the harness ingests directly ($--specularity-ipy$), scored against a future gridded intrusion survey. We thus state variable- ϕ RTN as a falsifiable screen whose single open step is an independent connectivity observable, not a verified forecast.

3.5 The intrusion residence number Ro — thinning-paced vs hydraulic-limited . The $RTN=1$ surface is a *threshold*; the *pace* at which the intrusion front advances is a separate, testable quantity. The level-set advance speed is $v_{kin} = (dH/dt)/|\nabla m| = A \cdot dH/dt$ (geometric amplification $A = 1/|\nabla m|$); the newly near-flotation bed must then re-pressurise to ocean head over a hydraulic residence time τ_{hyd} (the §5 cavity $\leftrightarrow$ channel band, 0.01–2 yr), so the front moves at $v_{obs} = \min(v_{kin}, \ell/\tau_{hyd})$ and the **residence number** $Ro = v_{kin} \cdot \tau_{hyd}/\ell$ has a regime boundary at the critical residence $\tau_{crit} = \ell/v_{kin}$ ($Ro = 1$ exactly at $\tau_{hyd} = \tau_{crit}$). For the runaway-tail cells ($A = 0.70\text{km}/\text{m}$, $dH/dt = 1.5\text{m}/\text{yr}$): $v_{kin} = 1.05\text{km}/\text{yr}$, $\tau_{crit} \approx 1.9\text{yr}$, so $\sim 98\%$ of the residence band lies below $\tau_{crit} \Rightarrow$ intrusion is predicted **thinning-paced** ($Ro \approx 1$), turning hydraulic-limited only at the slow (~ 2 yr) end. The implied hydraulic diffusivity $D_{hyd} = \ell^2/\tau_{hyd} \approx 0.06\text{--}12.7\text{m}^2/\text{s}$ spans the distributed $\rightarrow$ channelized range. Falsifier: a DInSAR v_{obs} over the runaway cells with $Ro \gg 1$

would place $\tau_{hyd} > \tau_{crit}$ (hydraulic limitation); $Ro \approx 1$ confirms thinning-pacing (validation/synthetic/intrusion_residence_number.py, 7/7 tests). (Werder et al. 2013; Hewitt 2013)

4. The thermal sliding-lag kernel, falsified

4.1 The hypothesis (a motivated null). No established sliding theory predicts that surge lag equals the *full-thickness* ice thermal-diffusion time; we state $\tau_{ice} = H^2/\kappa_{ice}$ ($\kappa_{ice} = 1.09 \times 10^{-6} m^2 s^{-1}$) as an explicit, falsifiable straw hypothesis precisely so the data can reject it.

4.2 The test. On real Bedmap2 thickness at the 131 Siegfried & Fricker (2018) catalogued active lakes (130 with valid thickness; median $H = 2282m$, range 637–3905 m):

- $\tau_{ice} = H^2/\kappa_{ice}$: **median $\approx 1.5 \times 10^5$ yr** (p5 $\approx 2 \times 10^4$; p95 $\approx 3 \times 10^5$ yr).
- Observed post-drainage surge lags: **0.02–2 yr** (Stearns et al. 2008; Siegfried et al. 2016).
- $\Rightarrow$ the literal kernel is $\sim 8 \times 10^4 \times$ **too slow** at the median lake; the $\log_{10}\tau_{ice}$ histogram (10^4 – 10^6 yr) is fully disjoint from the observed band.

4.3 Robust to the thickness dataset. Recomputing on an independent ice-thickness dataset (USAP-DC 601470, Stubblefield et al. 2021; ice thickness after Arthern et al. 2015): median thickness 2356 m, τ_{ice} median $\approx 1.55 \times 10^5$ yr; the 601470-vs-Bedmap2 thickness at the same centroids agrees at $r = 0.970$, so the falsification is not an artefact of one sampling.

4.4 Why the falsification is diagnostic. Two facts exclude ice thermal diffusion as the lag-setting mechanism: (i) the thermal skin depth at surge periods is $\delta_{skin} = \sqrt{\kappa_{ice}P/\pi} \approx 0.5$ – $5m$ ($P = 0.02$ – $2yr$) — the perturbation never penetrates beyond metres of ice, so the full-thickness H is the wrong scale; (ii) a drainage is an *impulse*, whose semi-infinite diffusive response is a monotone $t^{-1/2}$ decay with **no peak**, yet observed speed-ups **rise to a peak**. The lag is therefore **hydromechanical**, not thermal — the memory relocates into subglacial hydrology (cavity/channel pressurisation, effective-pressure control of sliding), with the ice-thermal $t^{-1/2}$ term subdominant.

5. The replacement kernel is an exact Mori–Zwanzig memory kernel

We refine the ontology to **two driving potentials** — the Leray pressure constraint (elliptic) and temperature (parabolic) — plus a **distinct, nonlinear-parabolic hydraulic subsystem** (cavity $\leftrightarrow$ channel storage–transport, with ϕ the hydraulic potential, *not* the Leray pressure). Its lumped linearisation $\dot{x} = Mx + F$, $x = (s, q)$ (s the resolved cavity store, q the eliminated channel variable, $M = [[-1/\tau_1, -a], [b, -1/\tau_2]]$), reduces by exact projection (linear Nakajima–Zwanzig / Mori, no Markov approximation) to

$$\dot{s}(t) = M_{ss}s(t) + \int_0^t K(t-\tau)s(\tau)d\tau + R(t), \quad K(\tau) = M_{sq}M_{qs}e^{M_{qq}\tau},$$

a non-local (memory) equation whose kernel $K(\tau) = -ab \cdot e^{-\tau/\tau_2}$ is exactly the channel subsystem’s own Green’s function weighted by the up/down couplings. The harness reuses the **same coupled**

model as the lag-shape/lag-value derivations and certifies five properties (validation/synthetic/hydraulic_mz_projection_synthetic.py, 6/6 tests):

1. **Projection exact to machine precision.** The GLE Laplace transfer function equals the full 2×2 resolvent's (1,1) component for 400 random stable overdamped systems — max abs err 5.0×10^{-16} .
2. **The kernel IS the eliminated subsystem's Green's function** — reproduced to 6.9×10^{-18} , decaying at exactly $1/\tau_2$, sign-definite.
3. **The reduced GLE reproduces the resolved trajectory** to rel-err 9.1×10^{-8} over the transient.
4. **Memory is necessary for the lag/peak.** The full response peaks at the interior time set by the coupled eigenvalues ($t^* = 0.911$, analytic 0.911); the Markovian adiabatic-elimination closure collapses to a monotone response with argmax at $t = 0$.
5. **Markovian limit = adiabatic elimination (DC gain).** $\int_0^\infty K d\tau = M_{sq}M_{qs}/(-M_{qq})$ (err 7.5×10^{-9}); shrinking the channel time $\tau_2 \rightarrow 0$ at fixed DC gain collapses the kernel toward $(\int K) \cdot \delta(\tau)$, recovering the local closure — the FDT-Markov limit, here *derived from the projection* rather than assumed.

This certifies the *structure* of the replacement mechanism (that the surge-lag memory is a genuine MZ kernel obtained by projecting out the channel, and how it degenerates to a local closure). (Werder et al. 2013; Hewitt 2013; Schoof 2010; Röthlisberger 1972)

Honest scope. This validates the **projection structure** of the *linearised lumped* model. It does **not** validate the physical identity of ϕ as distinct from the Leray pressure, that this subsystem dominates a real surge, or any field lag value — those remain, gated on real co-temporal velocity data (§6).

5.3 Physical magnitudes — the dimensional bridge and the roughness closure. The reduced kernel is dimensionless; the bridge to metres uses only known ice constants. With $\rho_i L = 3.0 \times 10^8 Jm^{-3}$, Glen A , effective pressure N , and literature subglacial inputs (Q , $\partial\phi/\partial s$), the steady R-channel $S^* = V_o/k_{creep}$, $R^* = \sqrt{2S^*/\pi}$, $\tau = 1/k_{creep}$ is **metre-scale** (central $R^* \approx 2.4m$, $\tau \approx 0.18yr$; 73 % of the literature box gives 0.1–50 m) — the R-channel regime, no free fudge factor; the wide upper tail is the low- N near-flotation limit ($k_{creep} \propto N^3 \rightarrow 0$, the §3 fold reappearing in the hydrology). Crucially $\rho_i L$ **cancels** in the scallop fraction $\Delta S/S = V_{scallop}/V_o = 0.33$, so a scalloped reach grows channels **+33 % in area** (calibration-free); the only genuine calibration is the concentration gain $g \in [0.1, 0.9]$ (network competition, direction- invariant). The roughness leg closes the ϕ loop: $z_0 = c_z \cdot a$ feeds the log-law drag $C_d = [\kappa/\ln(H/z_0)]^2$, and although the Nikuradse prefactor c_z is $\sim 10\times$ uncertain, the log law **buffers** it — at the Curl anchor a $10\times$ z_0 uncertainty propagates to only $\sim 1.7\times$ in C_d (and $\partial\phi/\partial s$), the least damaging open closure (validation/synthetic/a3_dimensional_bridge.py, a2_z0_roughness.py; 6+5 tests). (Röthlisberger 1972; Werder et al. 2013; Cuffey & Paterson 2010; Nikuradse 1933; Curl 1966)

6. The matched lag test — an honest null

Both halves of a matched test are open: drainage **dates** from CryoSat-2 in-lake elevation (*detect_{drainages}* extracts 7 fill→drain events across 5 lakes, including the documented Mercer ≈ 2012.5 drainage),

and velocity **response** from ITS_LIVE box-mean surface speed.

lake	base speed (m/yr)	drainages	annual noise floor	quarterly scatter	peak post-drainage
Mac1	422	3	1.0%	2.3%	+0.56σ
Mac2	402	1	1.1%	2.1%	+0.27 σ
Mac3	394	1	1.1%	3.4%	-0.12 σ

Result. No matched lake shows a post-drainage surge above the 2σ floor (peak $+0.56\sigma$, 0/5 significant). The lag value stays, *not* promoted to.

Why null is a *stronger* statement than “too noisy”. The modern ITS_LIVE record is well resolved (annual $\approx 1\%$, quarterly $\approx 2-3\%$), so the limit is not a noise budget. Two distinct, quantifiable limits remain: (1) **temporal aliasing** — the derived lag is sub-annual (t^* baseline ≈ 0.01 yr, p95 ≈ 0.1 yr), finer than the finest *robust* ITS_LIVE bin (quarterly, 0.25 yr); a brief transient is averaged out regardless of scatter (a Nyquist limit); (2) **amplitude bound** — a *sustained* speed-up would survive aliasing as a step, yet the peak anomaly is only $+0.56\sigma$, bounding any sustained surge to $\lesssim 10$ m/yr on ~ 400 m/yr. These map onto the two routes to: **sub-annual GPS/GNSS** (EarthScope/POLENET; beats aliasing) **or fast-outlet trunk velocity** (e.g. Byrd, Stearns et al. 2008; steep $v(N)$ sensitivity beats the amplitude floor), paired with that site’s drainage dates. The thermal H^2/κ kernel stays regardless.

Vetted forcing obtained. USAP-DC 601439 (Smith et al. 2009 ICESat) yields **58 drainage events across 52 lakes** (median drained volume 0.128 km^3 , max 2.92 km^3 ; implied water flux q_{water} median $\approx 2.5 \text{ m}^3/\text{s}$, p95 $\approx 31 \text{ m}^3/\text{s}$ — squarely in the literature subglacial-flood range), anchoring the §4/§5 forcing magnitude to real observations.

7. A field-measurable sliding law: the $s_N(N)$ master curve, three probes, and an ungrounding early-warning

The §4 falsification relocated the surge memory into effective pressure N . This section turns N from a tuned coefficient into a **measured** quantity with a closed-form, invertible sliding law and an operational precursor — the constructive capstone of the paper’s sliding-law arc (validation/synthetic/sn_master_curve.py, tidal_admittance_probe.py, spatial_ews.py; 17 tests).

7.1 $s_N(N)$ as the common observable. A lake drainage is an in-situ effective- pressure *step* whose surge amplitude $du/u = |s_N| \cdot (dN/N)$ measures the sliding sensitivity $s_N = d\ln u_b/d\ln N$; ocean thermal forcing lowers N *continuously*, so the gating slope $d\ln u_*/dT$ measures the **same** s_N ($+0.46 \rightarrow +0.62/^\circ\text{C}$, direct BedMachine N); ocean tides oscillate N at known amplitude, a third probe (7.3). All three read one curve.

7.2 The closed-form master curve. Solving the regularized-Coulomb law $\tau_b = CN(u/(u + u_0))^{1/m}$ at $\tau_b = \tau_d$ gives, exactly, $|s_N|(N) = m/(1 - (N_c/N)^m)$, $N_c = \tau_d/C \rightarrow m$ far from flotation, a **simple pole** at N_c ($\approx N_c/(N - N_c)$). Verified to 1.4×10^{-4} against the numeric s_N . This *derives* the near-flotation weakening that Joughin, Smith & Schoof (2019) impose by hand (ad hoc $h_{af} < h_T$ ramp + fixed u_0).

7.3 Inversion — measuring N_c and m . A drainage-step population recovers the flotation/Type-III threshold N_c to **0.2 / 0.4 / 1.0 %** at 5 / 10 / 20 % amplitude noise; m is degenerate with the per-event drop far from flotation. Ocean **tides** give a continuous third probe: the fundamental velocity admittance equals $|s_N|$ and the 2f/1f harmonic ratio diverges toward N_c , so admittance + harmonics + the known tidal amplitude recover m and the dimensionless flotation proximity $R = (N_c/N)^m$ **from surface velocity alone** — a sliding-law reading of the Gudmundsson non-linear MSf response, answering the “no knowledge of N ” objection. The tidal forcing amplitude is **measured**, not assumed: sampling the CATS2008 tide model (USAP-DC 601235) across the Bed-Machine grounding zone (within 10 km of the line; $n = 1108$ valid cells) gives a tidal ocean-pressure swing with median **0.30 % of ice overburden** (median tide $\eta \approx 0.84m$), rising monotonically from 0.21 % at 8–10 km to 0.45 % at 2–4 km from the grounding line (`validation/reports/tidal_forcing_gz.json`).

7.4 An ungrounding early-warning (operationalizing established CSD theory). The N_c fold makes the basal drag stiffness $\partial\tau_b/\partial u \propto (1 - R)^2/R \rightarrow 0$ (critical slowing down), so an ice stream approaching ungrounding should show **rising variance and lag-1 autocorrelation of its surface speed** (Kendall- $\tau \approx 0.54$ in an OU demonstration with N declining toward N_c). Critical-slowness-down early-warning is established (Scheffer et al. 2009; Dakos et al. 2008) and has already been applied to marine-ice-sheet tipping — Rosier et al. (2021) detected rising lag-1 autocorrelation and variance in grounding-line flux ahead of Pine Island tipping, and Boers & Rypdal (2021) in Greenland surface melt. The contribution here is not the early-warning concept but a specific **operational route**: reading $R(t) = (N_c/N)^m \rightarrow 1$ from the continuous tidal admittance of surface velocity. A single-snapshot **spatial** analogue (variance + along-flow correlation length rising toward the grounding line) is also derived (exact Lyapunov solve).

7.5 The framework on the same real lakes. Reading the §6 matched data (open CryoSat-2 drainage + ITS_LIVE velocity, 3 MacAyeal lakes) through this lens (`validation/external/lake_lag_sn_ews.py`, 5 tests): across the 3 resolved drainage events $|\Delta v/v| \leq 2\%$, mixed sign, near the ~ 1 % year-to-year noise floor — **0 positive $>2\sigma$ surge detections** — so via the master curve these trunk lakes sit **far from the N_c pole**; and **no** lake shows the joint critical-slowness-down signature (rising variance *and* high AC1), so the early-warning correctly returns a **true negative** on stable ice (unit tests confirm the detectors *do* fire on planted surges/CSD, so the null is real). Both probes agree: far from ungrounding — consistent with the 1/19 population non-detection.

7.6 Scope and falsification. Absolute s_N is uncalibrated (per-event dN/N is a lumped-storage lower bound); the robust claims are the sign/shape and the *threshold* N_c . With $n=8$ annual points per lake the CSD test is weak; a strong test needs the dense quarterly/GPS series and a stream actually nearing flotation. Falsifiers: a co-located dN +GPS population whose $|s_N(N)|$ does not follow $m/(1 - (N_c/N)^m)$; a stream nearing flotation with no variance/AC1 rise; tidal admittance/harmonics not steepening toward the grounding line. (Schoof 2005; Gagliardini et al. 2007; Joughin, Smith & Schoof 2019; Tulaczyk et al. 2000; Gudmundsson 2011; Scheffer et al. 2009; Dakos et al. 2008; Boers & Rypdal 2021)

8. Estimator calibration (synthetic plant-and-recover)

Every real-data estimator has a synthetic null/recovery harness exercised in `pytest`:

- **RTN classifier** — plant H^* , recover $RTN > 1$ region ($rtn_{synthetic}$).

- **Lag estimator is kernel-shape-generic** — plant one target lag with five markedly different causal kernels (Gamma $k = 2/4$, log-normal, bi-exponential, raised-cosine); $estimate_{lag}$ recovers it to $\leq 10\%$ for every shape, delta-kernel control returns 0. So a recovered lag is a property of the data, not the assumed Gamma form.
- **Grounding-line migration level-set law is exact** — $|dH/dt|/|\nabla m|$ reproduced to rel-err 5×10^{-16} (1-D) / 4×10^{-14} (tilted 2-D); the thinning-paced null is flat by construction; a planted u_* discount is recovered sign-faithfully and is unbiased under 25 % noise / null under permutation.
- **ϕ -area inversion, MISI band, intrusion-clock drivers** — calibrated on synthetic ground truth.
- **CMN commutator identity** — to $\sim 10^{-7}$ ($cmn_{synthetic}$).

These guarantee the §3–§6 verdicts are read against the right null with unbiased, sign-faithful estimators.

9. Discussion and limitations

What this is. A machine-precision certification of the cavity $\leftrightarrow$ channel kernel’s exact MZ structure (the central positive result); a motivated null on a literal thermal sliding kernel; an honest matched-lag null with a precise diagnosis and a concrete unblock path; a flotation-ratio intrusion *screen* with an independent cross-check; and a field-measurable sliding law (the $s_N(N)$ master curve, an N_c inversion, and an early-warning that operationalizes established CSD theory) read honestly against the same real lakes.

What this is not. No RTN precision/recall (no gridded survey). No verified surge-lag *value* (the matched test is null; the value is to order of magnitude only). No nonlinear hydraulic-kernel derivation (the §5 result is the *linearised lumped* projection; a coupled GlaDS/Röthlisberger derivation is future work). The physical distinctness of ϕ from the Leray pressure remains unproven.

The open gates, each with a concrete unblock.

1. **Matched lag** $\rightarrow$: sub-annual GPS/GNSS (EarthScope/POLNET) or fast-outlet trunk velocity, paired with drainage dates.
 2. **Derived nonlinear hydraulic kernel:** a coupled GlaDS/Röthlisberger cavity $\leftrightarrow$ channel model (replace the lumped RC analogue).
 3. **RTN precision/recall:** a future gridded intrusion survey (NISAR / IceBridge successors) to score against.
 4. **s_N law / early-warning** $\rightarrow$: a co-located dN+GPS (or tidal- admittance) population at a stream nearing flotation, to fit $m/(1-(N_c/N)^m)$ and test the variance/AC1 rise; and a DInSAR v_{obs} over runaway intrusion cells to place Ro on the thinning-paced vs hydraulic-limited curve (§3.5).
-

10. Reproduce

```
pip install -r requirements.txt
python glaciers/validation/external/run_rtn_bedmap2.py --stride 3 --phi 0.9 # §3 RTN (Fig 57 family)
# §3.3 BedMachine cross-check: glaciers/validation/external/rtn_corollaries_bedmachine.py (GPU)
python glaciers/validation/external/rtn_baseline_skill.py --stride 5 # §3.4
↪ RTN==thickness-above-flotation (zero skill at constant phi)
python glaciers/validation/synthetic/rtn_variable_phi_skill.py # §3.4b variable-phi is the only escape
↪ (plant-and-recover, 5 tests)
python glaciers/validation/external/rtn_variable_phi_real.py --stride 10 # §3.4b hydraulic-routing
↪ phi(x,y) reorders RTN on real Bedmap2
python glaciers/validation/external/run_sliding_real.py --with-velocity Mac1 # §4  $\tau_{ice}$  + ITS_LIVE
python -m glaciers.validation.external.lag_fit_real # §6 matched-lag null
python glaciers/validation/external/run_usapdc_lakes.py # §6 vetted forcing (58 events)
pytest glaciers/tests/test_validation_synthetic.py -v # §8 estimator calibration + §5 MZ projection
↪ (6/6)
python glaciers/validation/synthetic/intrusion_residence_number.py # §3.5 intrusion residence number
↪  $R_o$ 
python glaciers/validation/synthetic/sn_master_curve.py # §7  $s_N(N)$  master curve + inversion
python glaciers/validation/synthetic/tidal_admittance_probe.py # §7.3 tidal third probe
python glaciers/validation/external/lake_lag_sn_ews.py # §7.5 §I framework on real lakes
```

Data and code availability

The intrusion result (Figs. 1–2) uses open datasets — Bedmap2, BedMachine Antarctica v4, CryoSat-2 lake-drainage dates, and ITS_LIVE velocities — and regenerates from the scripts in the reproduce section. All analysis code and derived products are in the public repository at <https://github.com/abdh0saifelden2-debug/K>.

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